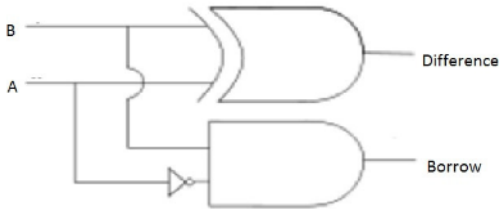
Circuit Simulation Lab: 3

The purpose of this experiment is to introduce the design of simple combinational circuits, in this case half subtractor, and full subtractor



Half Subtractor

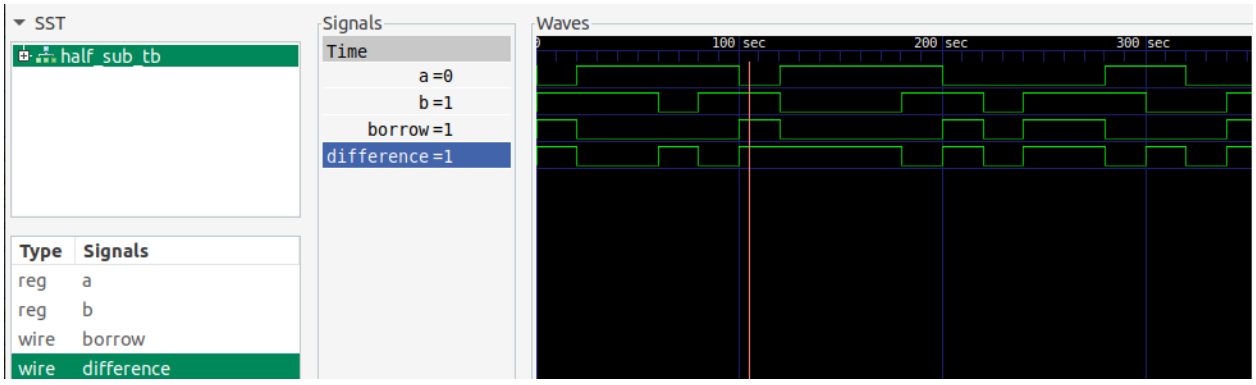
Dataflow modeling
//dataflow
module half_sub (
input a , b ,
output borrow , difference);

```
assign difference = a ^ b ;
assign borrow = ~a & b ;
```
endmodule

Tb
module half_sub_tb ;
reg a , b ;
wire borrow , differnce ;

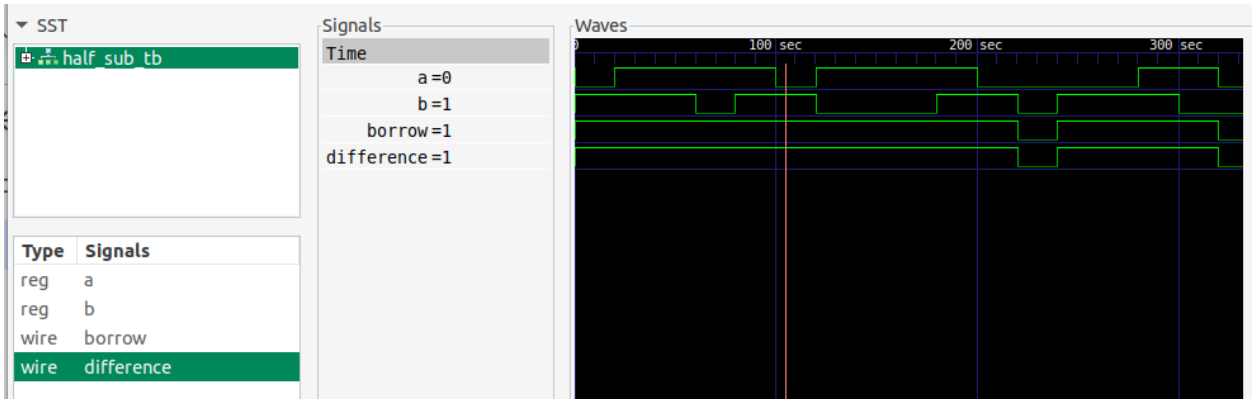
half_sub h1 (.a(a),.b(b),.borrow(borrow),.difference(difference));

initial begin
$dumpfile ("half_sub.vcd");
$dumpvars(0,half_sub_tb);
repeat (20) begin
a = $random %2 ;
b = $random %2;
#20;
$display ("a = %d , b = %d , borrow = %d , difference = %d " , a,b,borrow,difference);
end
$finish;
end
endmodule



Behavior modeling
//behavioural modeling (BM)
module half_sub_BM(
input a , b ,
output reg borrow ,difference);

always @(a | b) begin
borrow = ~a & b;
difference = a ^ b ;
end
endmodule

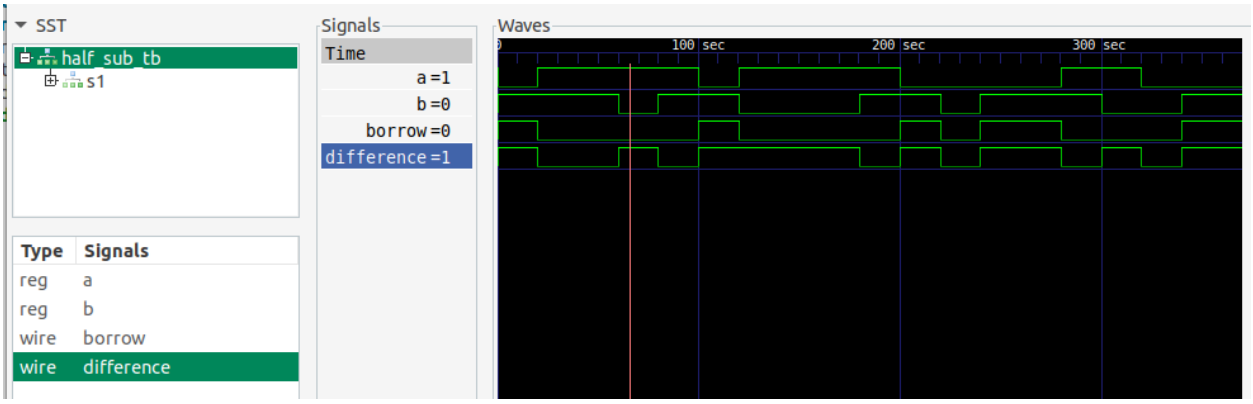


Structural modeling
XOR + AND +NOT = half sub

module xor1(
input a , b,
output c);

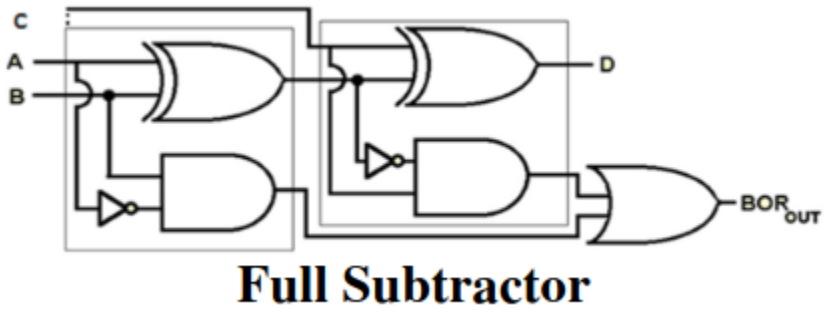
```
assign c = a ^ b ;
endmodule

module not1 (
input a ,
output not_a);

assign not_a = ~ a;
endmodule

module and1(
input a , b ,
output c );

assign c = a & b ;
endmodule

//structural modeling
module half_sub_SM(
input a , b ,
output borrow ,difference);
wire not_a;
xor1 x1 (.a(a),.b(b),.c(difference));
not1 n1 (.a(a),.not_a(not_a));
and1 a1 (.a(not_a),.b(b),.c(borrow));
endmodule

module half_sub_tb ;
reg a , b ;
wire borrow , differnce ;

half_sub_SM s1 (.a(a),.b(b),.borrow(borrow),.difference(difference));

initial begin
$dumpfile ("half_sub_SM.vcd");
$dumpvars(0,half_sub_tb);
repeat (20) begin
a = $random %2 ;
b = $random %2;
#20;
$display ("a = %d , b = %d , borrow = %d , difference = %d " , a,b,borrow,difference);
end
$finish;
end
```

endmodule




```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_3$ iverilog -o out x
or1.v not1.v and1.v half_sub_SM.v half_sub_tb.v
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_3$ vvp out
VCD info: dumpfile half_sub_SM.vcd opened for output.
a = 0 , b = 1 , borrow = 1 , difference = 1
a = 1 , b = 1 , borrow = 0 , difference = 0
a = 1 , b = 1 , borrow = 0 , difference = 0
a = 1 , b = 0 , borrow = 0 , difference = 1
a = 1 , b = 1 , borrow = 0 , difference = 0
a = 0 , b = 1 , borrow = 1 , difference = 1
a = 1 , b = 0 , borrow = 0 , difference = 1
a = 1 , b = 0 , borrow = 0 , difference = 1
a = 1 , b = 0 , borrow = 0 , difference = 1
a = 1 , b = 1 , borrow = 0 , difference = 0
a = 0 , b = 1 , borrow = 1 , difference = 1
a = 0 , b = 0 , borrow = 0 , difference = 0
a = 0 , b = 1 , borrow = 1 , difference = 1
```

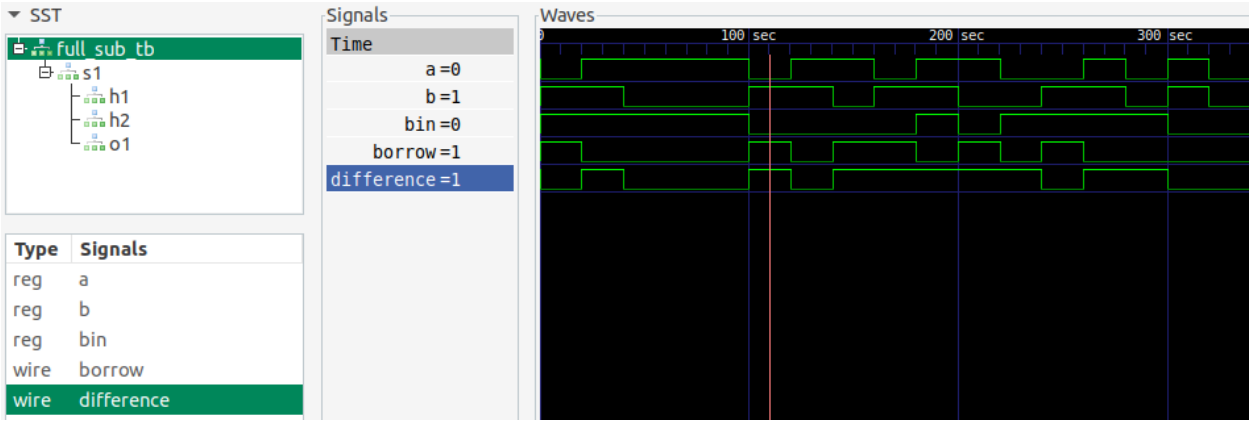
FULL ADER



Full Subtractor

Structural modeling
module full_sub_ST(
input a , b , bin ,
output difference , borrow);
wire h1_diff , h1_borrow;
wire h2_borrow;

```
half_sub h1 (.a(a),.b(b),.difference(h1_diff),.borrow(h1_borrow));
half_sub h2 (.a(bin),.b(h1_diff),.difference(difference),.borrow(h2_borrow));
or1 o1 (.a(h1_borrow),.b(h2_borrow),.out1(borrow));
Endmodule

//dataflow
module half_sub (
input a , b ,
output borrow , difference );

assign difference = a ^ b ;
assign borrow = ~a & b ;
endmodule

module or1(
input a , b ,
output out1);
assign out1= a | b ;
Endmodule

module full_sub_tb ;
reg a , b , bin;
wire borrow , differnce ;

full_sub_ST s1 (.a(a),.b(b),.bin(bin),.borrow(borrow),.difference(difference));

initial begin
$dumpfile ("full_sub_ST.vcd");
$dumpvars(0,full_sub_tb);
repeat (20) begin
a = $random %2 ;
b = $random %2;
bin = $random %2;
#20;
$display ("a = %d , b = %d ,bin = %d , borrow = %d , difference = %d " ,
a,b,bin,borrow,difference);
end
$finish;
end
endmodule
```




```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_3/full_subs$ iverilo
g -o out or1.v full_sub_tb.v full_sub_ST.v  half_sub.v
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_3/full_subs$ vvp out
VCD info: dumpfile full_sub_ST.vcd opened for output.
a = 0 , b = 1 ,bin = 1 , borrow = 1 , difference = 0
a = 1 , b = 1 ,bin = 1 , borrow = 0 , difference = 1
a = 1 , b = 0 ,bin = 1 , borrow = 0 , difference = 0
a = 1 , b = 0 ,bin = 1 , borrow = 0 , difference = 0
a = 1 , b = 0 ,bin = 1 , borrow = 0 , difference = 0
a = 0 , b = 1 ,bin = 0 , borrow = 1 , difference = 1
```